

Some Examples Related to Homework 01

1 Piano Tuners, Using Subscripts

How Many Piano Tuners are there in Chicago? the classic one!

Notation:

w = worker; keep t for tuning, not tuner...

use p for piano, not people

N_c = 3 million people in Chicago (...as opposed to metropolitan area)

$N_{c/h}$ = 4 Chicagoans per household

$N_{h/p}$ = 20 households per piano

$N_{t/p-y}$ = 1 tuning per 2 piano-years

$N_{d/y}$ = 250 days per year (at work)

$N_{t/w-d}$ = 4 tunings per worker-day

Note that we could often use the reciprocal of these factors, and alter the equation...

$$N_w = \left[N_c / (N_{c/h} N_{h/p}) \right] \left[N_{t/p-y} / (N_{d/y} N_{t/w-d}) \right]$$

calculate: $[(3000000 / (4*20))] [(1/2) / ((250 * 4))] = 19$

Some older solutions get larger numbers, due to more pianos per household.

Check units: “subscripts cancel”

Checks: First block is N_p , second is $N_{w/p}$ (workers *required* per piano)

Can also check “if input N_x increases, then the answer ...”

Q. What’s up with “worker-day”?

Can think of as 4 tunings/day/worker = 4 tunings/worker-day

Careful: this can get confusing sometimes...

Alternate method: “supply = demand” (from a student, 2023)

$$N_w N_{d/y} N_{t/w-d} = \left[N_c N_{t/p-y} / (N_{c/h} N_{h/p}) \right]$$

Both sides are $N_{t/y}$: available = needed.

2 Parametric Form for Uncertain Inputs

Number of cells in our bodies?

Sharpen: typical-size adult (would that matter? bio question...)

Strategy: ratio of cell volume to total volume

Total body volume:

75 kg at density of $\sim 1 \text{ g/cm}^3$ (water) $\rightarrow 7.5 \times 10^{-2} \text{ m}^3$

$$V = M/\rho = 75 \text{ kg} \left(\frac{1 \text{ cm}^3}{\text{g}} \right) \left(\frac{10^3 \text{ g}}{\text{kg}} \right) \left(\frac{1 \text{ m}^3}{10^6 \text{ cm}^3} \right) = 7.5 \times 10^{-2} \text{ m}^3$$

Total cell volume: need cell diameter D , then use $(\pi/6)D^3$

Typical cells can't be seen by eye, so less than 0.1 mm

Can be seen in light microscope, so larger than 500 nm wavelength

Answer is probably between 1 and 100 microns; take *geometric mean*

So, a diameter of $1 \times 10^{-5} \text{ m}$

Then $V_{\text{cell}} \simeq 5. \times 10^{-16} \text{ m}^3$

(Note: we could worry about inter-cellular fluids, or use a cube, or...)

Substituting, we get: 1.5×10^{14} cells, or about 150 trillion cells

Comment on parametric answers:

For any, or all, parameters we can explicitly display the dependence on our inputs

It's another example of "multiplying by one" (like we do to change units)

Here, $N_{\text{cell}} = k/D^3$ where k is everything but the diameter.

I write D_0 to represent our particular assumed value.

So, if we multiply by $(D_0/D_0)^3$ and re-arrange, we can cleverly separate a *nominal numerical answer* and functional dependence:

$$N_{\text{cell}} = \frac{k}{D^3} = \left(\frac{k}{D_0^3} \right) \left(\frac{D_0}{D} \right)^3 = 1.5 \times 10^{14} \left(\frac{10^{-5} \text{ m}}{D} \right)^3$$

WARNING: it's very easy to get the factors inverted! So check that it makes sense.

If we assumed a smaller D , the number of cells increases: OK

What is the advantage? Two nice things are combined:

- What is the answer for some typical value of D ? (i.e., $D = D_0$)
- What is the dependence? (i.e., $x^3, 1/x$, etc.)

We could do this parametric form for any factor (or > 1 factor) we like.

Aside on taking the mean above:

When we're dealing with strings of multiplied factors and large uncertainties, using the geometric mean is probably best.